

Raft Building

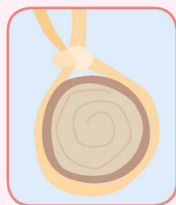
Does the ocean call to you? Take to the water like Moana with this raft project that combines science, technology, engineering and math. See how far you'll go with the wind in your sail!

You will need:

- Approximately 14 thin, straight twigs
- Plain fabric
- Yarn
- Pencil
- Scissors
- Fabric pen
- Clear tape
- Darning needle
- Strong glue

Make a raft

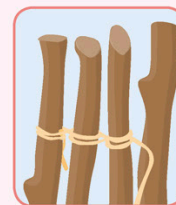
You'll need about 10 thin, straight twigs for the base. If they're not the same length, ask an adult to trim them for you. You'll also need two twigs for the sail and two shorter ones to cross the raft.



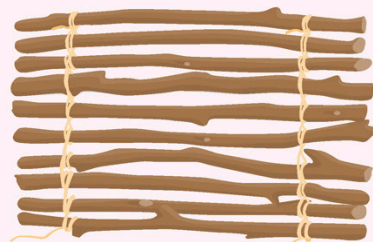
- 1** Tie a piece of yarn around the end of one twig.



- 2** Wind it around a second twig, making a figure-of-eight loop. Repeat this backwards and forwards a few times until it feels secure.



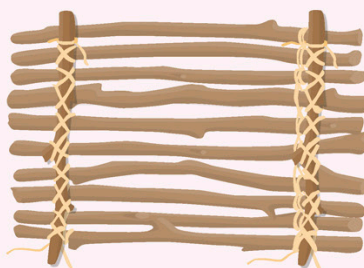
- 3** Repeat the same pattern from the second to a third twig and on until all 10 twigs are connected together.



Tie off and trim your loose ends.



This twig should be long enough to lie across your raft.



- 4** Tie a tight knot after the tenth twig and trim the excess yarn. Repeat this process for the other side of the raft, so both edges are secured.

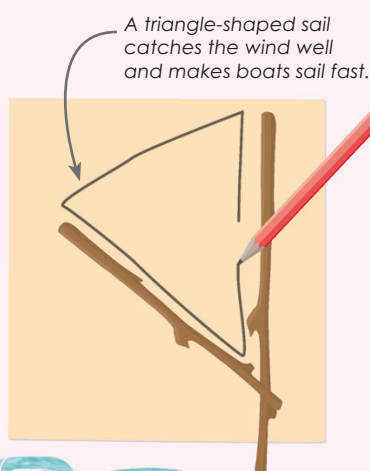
- 5** Tie yarn around one end of a shorter twig.

- 6** With the raft lying sideways, place the twig vertically across one end. Thread the yarn around the twig, weaving it in and out of the twigs underneath.

- 7** Follow the same process with a new twig at the other end of the raft.

Add a sail

You can't sail a boat without a sail! Moana's is decorated with the red, swirling symbol that was on the heart of Te Fiti.



A triangle-shaped sail catches the wind well and makes boats sail fast.



- 2** Draw between the twigs to make a triangular sail on the fabric. Cut it out.



- 3** Using a fabric pen, draw Moana's swirling symbol on the sail.



- 4** Wind red yarn around the two twigs to join them together in a figure-of-eight loop, as above.

TRY THIS!
To keep the twigs still while you weave, you could temporarily stick them to a table with clear tape.

